Format Over Correctness: Training Small LLMs to Solve Systems of Equations by Method

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1. Objective

Build a verified dataset of math problems paired with **multiple solution methods**, and test how to fine-tune a small language model to solve a linear system by a *specified* method (substitution, elimination, Cramer's rule, matrix inverse, Gaussian elimination, graphing). The controlled question: what moves accuracy — the base model's prior, the **format** of the training targets, the training stage (SFT vs. DPO), or data volume?

2. Data and verified pipeline

- **Source:** `CAHLR/OATutor-Content` (13,287 problems, 17,761 steps). The systems-of-equations slice has 751 problems; **164** state a parseable, uniquely-solvable linear system in text.
- **Targets are generated, questions are not.** Questions are taken verbatim from OATutor; every *solution* is computed with **sympy** and verified by substituting the answer back into the original equations. **No language model produces the maths, so no hallucinated arithmetic can enter the data.**
- **Two target formats:** *terse* (compact steps) and *worked* (same maths, every computation shown, e.g. $3(3y) - 4y = 5 \Rightarrow 5y = 5$).
- **Preference pairs (for DPO):** a verified-correct solution vs. a deterministically corrupted, verified-wrong one.
- **Verification (independent re-check): 0 failures** across 811 real and 5,624 synthetic method-solutions and all preference pairs.

3. Experimental setup

- **Split by system:** 164 → 114 train / 24 valid / **26 test** (126 method-solutions), no leakage, identical across all conditions.
- **Two matched base models (4-bit, MLX, Apple Silicon):** `Qwen2.5-1.5B-Instruct` (general) and `Qwen2.5-Math-1.5B-Instruct` (math).
- **Training:** LoRA (8 layers) via `mlx-lm-lora`; SFT then DPO ($\beta = 0.1$).
- **Metric:** exact answer accuracy on held-out systems, graded by sympy (no LLM judge).

4. Results

Both bases start at essentially the same accuracy (0.389 vs 0.381) but with different per-method strengths, giving a clean controlled comparison.

Model	Base	Terse SFT	Terse DPO	Worked SFT	Worked DPO	Scaled SFT
general	0.389	0.087	0.016	**0.159**	0.000	0.095
math	0.381	0.214	0.135	**0.286**	0.111	0.270

Table 1: Exact answer accuracy on the 26 held-out real systems (126 method-solutions).

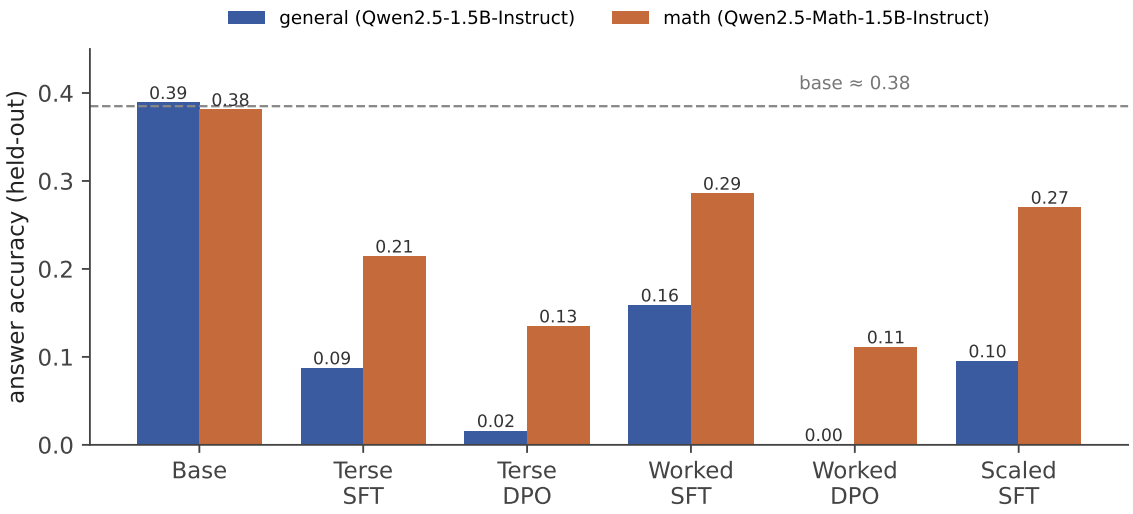


Figure 1: Overall accuracy by training condition. No trained variant beats the base overall; worked $\geq$ terse everywhere; DPO $\leq$ SFT everywhere.

Mechanism (why terse hurts). After terse SFT the model imitates the format perfectly (126/126 outputs end with a clean "Solution:") but stops reasoning and fabricates the intermediate step: on $2x - 6y = 0,\ 3x - 4y = 5$ (answer $x = 3, y = 1$) it writes $x = 3y$ then invents $15 - 18y = 0$ (should be $5y = 5$). Worked targets, which show that step, recover most of the loss (substitution: terse 0.12 $\rightarrow$ worked 0.44 for the general model).

5. Key findings

1. **Format > correctness of the target.** Worked targets beat terse at SFT for *both* models (general +8pt, math +7pt) — the largest controllable lever here.
2. **SFT $\gg$ DPO at 1.5B/4-bit.** Every DPO run scored below its SFT start (general collapsed to ≈ 0), despite perfectly separable preference pairs — a capacity boundary, not a data flaw.
3. **More data $\neq$ better.** Scaling 7$\times$ with verified synthetic systems left the math model flat and the general model worse. On inspection the synthetic set is 0% fractional-solution (vs 37% in real data) and the scaled run used more iterations — so **distribution and training budget matter more than raw volume.**
4. **Training helps where the base is weak.** No trained variant beats base *overall* (bases already reason well on known methods), but worked SFT **beats the base on methods it lacked** — math Cramer's rule 0.00 $\rightarrow$ 0.15.

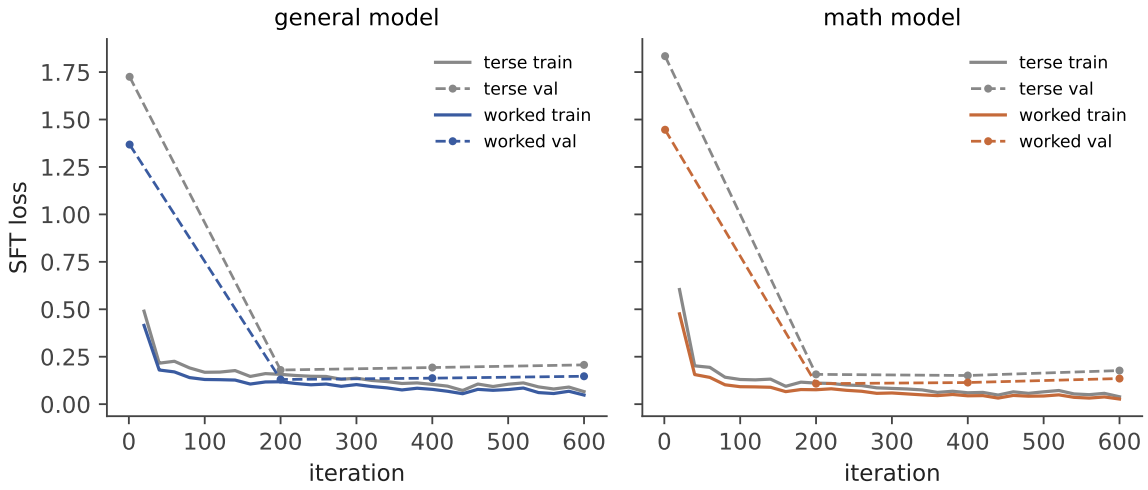


Figure 2: SFT training loss (train solid, validation dashed) for both models under terse vs. worked targets. Training converges cleanly in every case — the accuracy differences are not a training-stability artifact.

6. Limitations and next steps

- Small (1.5B / 4-bit) models on a laptop; the "training net-hurts a capable base" effect should shrink at 7B+ with more headroom.
- Clean scaling test needed: a **compute-matched** control (worked-114 at the scaled iteration count) to separate data volume from training length.
- Substantive fix: a **distribution-matched** synthetic set (~37% fractional solutions, matched coefficients), then re-scale.
- Report **method-faithfulness** (does it use the *requested* method) as a primary metric — logged, and where even the strong base has headroom.

Reproducible pipeline: questions verbatim from OATutor; solutions and grading by sympy; every target independently re-verified (0 failures). Repo: `Hassan-Raza-Shaikh/Preference-Learning-in-Education`.